Project Proposal – Group 15

Predicting LLM Response Quality Scores to Reduce Evaluation Costs

1. Business Problem

As large language models (LLMs) are rapidly deployed across enterprise applications — from customer service automation to financial report generation — ensuring the quality of model outputs has become a critical operational challenge. Today, organizations rely on two primary evaluation methods:

- Human annotation: Domain experts manually score LLM responses for accuracy, helpfulness, and coherence.
- LLM-as-Judge: A secondary, more powerful LLM (e.g., GPT-4) automatically scores the responses of the primary model.

Both approaches are expensive, slow, and do not scale. Human annotation can cost millions of dollars at scale, while LLM-as-Judge incurs significant API costs and introduces model-specific biases.

Our proposed solution is to build a supervised regression model that predicts human preference quality scores for LLM responses based on measurable textual features — without requiring a human evaluator or a secondary LLM at inference time. This approach delivers three forms of business value:

- **Cost savings:** Reduces dependence on expensive human labeling pipelines and API-based judge models.
- **Speed:** Enables near-real-time quality monitoring of LLM outputs in production.
- **Governance:** Provides an audit-ready, automated scoring mechanism aligned with AI risk frameworks such as NIST AI RMF and SR 11-7.

Crucially, our approach supports online inference: quality scores can be predicted in real-time for each generated response without requiring pairwise comparisons or external evaluation models. This makes the system well-suited for integration into live LLM deployment pipelines.

2. Prior Work: Existing ML Solutions

2.1 Reward Models in RLHF (Most Directly Relevant)

The most closely related prior work comes from research on **Reinforcement Learning from Human Feedback (RLHF)**, which trains reward models to predict scalar human preference scores from prompt-response pairs. Stiennon et al. (2020), *"Learning to Summarize from Human Feedback,"* and Ouyang et al. (2022), *"Training Language Models to Follow Instructions with Human Feedback" (InstructGPT),* both train regression-style reward models (typically fine-tuned transformers such as GPT or DeBERTa) to map a (prompt, response) pair to a continuous quality score, supervised by human preference labels.

At a high level, a reward model takes a tokenized prompt-response pair as input and outputs a single scalar score. It is trained using a ranking loss (e.g., Bradley-Terry) over pairs of responses where human annotators have indicated a preference. This formulation directly mirrors our regression setup.

Advantages: Directly predicts per-response quality scores from text; proven to correlate strongly with human preferences at scale; forms the backbone of production alignment systems at OpenAI and Anthropic.

Disadvantages: Requires large-scale human preference data to train; full fine-tuning of a transformer is computationally intensive; risk of reward hacking when the model is used to directly optimize another LLM.

Our project adopts a similar supervised regression framing but uses lightweight, interpretable features (lexical, semantic, structural) rather than end-to-end fine-tuning, making the approach feasible on a modern laptop.

2.2 Chatbot Arena and the Bradley-Terry Model

Zheng et al. (2023), *"Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena,"* introduced a crowdsourced platform where human users compare two LLM responses and vote for the preferred one. The **Bradley-Terry model** is then used to convert binary pairwise votes into a continuous Elo-style ranking score across models.

Advantages: Directly captures human preference signals; scales to millions of comparisons; produces interpretable model rankings.

Disadvantages: Produces aggregate model rankings only — not per-response quality scores; cannot be applied in real-time to individual new responses; requires an active comparison platform.

2.3 BERT Score: Semantic Similarity as a Quality Proxy

Zhang et al. (2019), *"BERTScore: Evaluating Text Generation with BERT,"* uses pre-trained BERT contextual embeddings to compute token-level cosine similarity between a generated response and a reference text, yielding Precision, Recall, and F1 quality scores.

Advantages: Captures semantic equivalence beyond n-gram overlap; correlates better with human judgments than BLEU/ROUGE; handles paraphrasing gracefully.

Disadvantages: Requires a reference text, which may not exist for open-ended tasks; does not directly model human preferences; computationally heavier than n-gram metrics.

In our project, BERTScore will be used as one feature input to the regression model, rather than as a standalone evaluation metric.

3. Datasets

3.1 LMSYS Chatbot Arena Conversations (Primary Dataset)

This dataset was released alongside the Chatbot Arena platform by Zheng et al. (2023) and is publicly available on Hugging Face.

- **Size:** ~33,000 crowd-sourced conversations with human preference labels
- **Coverage:** Responses from 20+ LLMs including GPT-4, Claude-v1, LLaMA, Vicuna, and others
- **Labels:** Human pairwise preference votes (model_a wins / model_b wins / tie), convertible to per-response win-rate regression targets
- **Additional fields:** Model names, conversation turns, timestamps, language tags
- **Access:** https://huggingface.co/datasets/lmsys/chatbot_arena_conversations

The pairwise vote structure will be transformed into a continuous win-rate score per response (i.e., the proportion of comparisons in which a response was preferred), creating a well-defined regression target suitable for model training and evaluation.

3.2 LMSYS-Chat-1M (Secondary / Feature Engineering Dataset)

As a supplementary resource, we will reference the LMSYS-Chat-1M dataset, which contains approximately 1 million real-world conversations with 25 LLMs.

- **Size:** ~1 million conversations across 25 LLMs
- **Coverage:** Collected from 210,000+ unique IP addresses across 154 countries
- **Additional fields:** Language, OpenAI moderation scores, conversation length, user turn count
- **Access:** https://huggingface.co/datasets/lmsys/lmsys-chat-1m

While this dataset does not include human preference labels, it provides a large-scale corpus for feature engineering analysis — helping us understand distributions of response length, structure, and toxicity signals that inform our feature extraction pipeline.

4. Feature Engineering and Evaluation Plan

4.1 Feature Engineering

We will extract the following feature categories from each LLM response to serve as inputs to the regression model:

Feature Type	Examples	Rationale
Lexical	Response length, avg word length, type-token ratio (TTR)	Captures verbosity and vocabulary richness
Semantic	BERTScore F1, sentence embedding cosine similarity	Measures meaning alignment and coherence
Structural	Bullet points, code blocks, headers, paragraph count	Reflects formatting quality and organization
Quality Signals	Perplexity (GPT-2), toxicity score, grammar error rate	Proxy metrics correlated with human preferences
Prompt-level	Prompt length, question type, topic category	Contextualizes response quality relative to task

Notably, we will include perplexity scores computed via GPT-2 as a proxy for fluency, toxicity scores from the Perspective API as a proxy for safety, and grammar error rates from language tool libraries — features that go beyond simple surface statistics and have been shown to correlate with human quality judgments in prior reward model literature.

4.2 Evaluation Metrics

We will evaluate model performance using the following metrics on a held-out test set:

Metric	Description	Purpose
RMSE / MAE	Root mean squared / mean absolute error on held-out set	Measures regression accuracy of predicted scores
Spearman Correlation	Rank correlation between predicted and human scores	Checks alignment with human preference rankings
Top-k Ranking Accuracy	% of top-k responses correctly identified by model	Practical evaluation for deployment scenarios

Primary evaluation: RMSE for regression accuracy and Spearman correlation to measure alignment with human preference rankings. Spearman correlation is particularly important as it directly captures how well our model preserves the relative ordering of response quality — the key signal underlying human preference data.

References: Ouyang et al. (2022). Training language models to follow instructions with human feedback. NeurIPS 2022. | Zheng et al. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. NeurIPS 2023. | Zhang et al. (2019). BERTScore: Evaluating Text Generation with BERT. ICLR 2020. | Stiennon et al. (2020). Learning to Summarize from Human Feedback. NeurIPS 2020.